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Immigration and Its Approximation

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Continuous-State Branching Process With Immigration and Its Discrete Approximation

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Mathematics and Applied Mathematics

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[ABSTRACT]: This thesis investigates the relationship between Galton–Watson processes and their continuous scaling limits. In the finite-variance regime, Feller’s scaling theorem establishes that, after appropriate normalization, Galton–Watson processes admit a continuous-state branching process (CSBP) as a limit, whose evolution is governed by Brownian motion.

When the offspring distribution exhibits heavy-tailed behavior, the classical Central Limit Theorem no longer applies and is replaced by stable limit theorems. In this setting, the limiting noise is described by an α -stable Lévy process, which naturally leads to a stable-driven model of the form

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t.$$

Such a process, however, does not in general satisfy the branching property.

To address this issue, we consider an extended model incorporating a drift term,

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t + c_t dt,$$

and study the conditions under which self-similarity is preserved. In particular, we determine the admissible form of c_t and analyze the corresponding Laplace transform structure.

Furthermore, based on Le Gall’s construction, we study the discrete-to-continuum scaling from Galton–Watson processes to CSBPs, and explain how the drift, Brownian, and jump components in the Lévy–Khintchine branching mechanism arise from discrete branching structures. We also characterize the asymptotic behavior of branching and immigration parameters leading to stable continuous-state branching processes with immigration (CBIs).

[Key words]: CSBP; Stable Lévy process; Self-similarity; Immigration; Laplace

transform.

[摘要]: 本文研究 Galton–Watson 分支过程与其连续极限之间的联系。在有限方差情形下, Feller 缩放极限定理表明, 经过适当归一化后, Galton–Watson 过程以连续状态分支过程 (continuous-state branching process, CSBP) 作为极限, 其演化由布朗运动刻画。

当后代分布呈现重尾行为时, 经典中心极限定理不再适用, 而由稳定极限定理所取代。在这种情形下, 极限噪声由 α -稳定 Lévy 过程描述, 从而得到如下稳定驱动的候选模型:

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t.$$

然而, 该过程一般不满足分支性质。

为此, 本文考虑引入漂移项的扩展模型:

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t + c_t dt,$$

并研究在何种条件下该过程仍保持自相似性。特别地, 本文推导了函数 c_t 的具体形式, 并分析了对应的 Laplace 变换结构。

此外, 本文基于 Le Gall 的构造方法, 研究离散 Galton–Watson 过程向连续状态分支过程的缩放极限, 说明 Lévy–Khintchine 型分支机制中的漂移项、Brownian 项与跳跃项如何由离散分支机制产生, 并刻画分支参数与移民项 (immigration) 的渐近行为, 从而得到稳定型带移民的连续状态分支过程 (continuous-state branching process with immigration, CBI)。

[关键词]: 连续状态分支过程; 稳定 Lévy 过程; 自相似性; 移民; Laplace 变换。

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1. Introduction

A fundamental starting point of this thesis is the classical result that the **Feller branching diffusion** arises as a continuous scaling limit of Galton–Watson branching processes, and is closely related to the 0-dimensional squared Bessel process.

In the finite-variance setting, the emergence of Brownian noise can be understood as a consequence of the **Central Limit Theorem (CLT)**: under suitable normalization, the aggregated fluctuations of the offspring distribution converge to a Gaussian limit, leading to a diffusion-type branching model.

A natural question is how this limiting structure changes when **the offspring distribution no longer has finite variance and instead exhibits heavy-tailed behavior**.

In such a regime, if the offspring distribution belongs to the domain of attraction of a stable law, the classical CLT is replaced by a **stable limit theorem**. Consequently, the Gaussian fluctuation is expected to be replaced by an α -stable Lévy fluctuation. This leads heuristically to a stable-driven stochastic model of the form

$$dY_t = a Y_{t-}^{1/\alpha} d\beta_t,$$

where β_t is a spectrally positive α -stable Lévy process.

As in the Feller case, the process formally preserves a **self-similar** scaling relation, with self-similarity index

$$H = \frac{1}{\alpha - 1}.$$

However, unlike the classical Brownian branching diffusion, such a stable-driven model does not automatically satisfy the **branching property**, and therefore does not necessarily define a continuous-state branching process (CSBP).

Motivated by this observation, we consider the more general stable-driven model

$$dY_t = a Y_{t-}^{1/\alpha} d\beta_t + c_t dt,$$

where the additional deterministic term $c_t dt$ is interpreted as a possible immigration mechanism.

The first objective of this thesis is to study, at a preliminary level, the structural properties of this model. In particular, we investigate the **condition on the function c_t under which the process preserves the same self-similar scaling** as the stable branching part. We also analyze the corresponding **Laplace transform structure through the generator**, motivated by the classical exponential-affine characterization of CSBPs.

A second objective is to **relate these continuous models back to their underlying discrete**

branching origins. Following the construction of Le Gall, we study how suitably rescaled continuous-time Galton–Watson processes converge to continuous-state branching processes. The key point is that the limiting branching mechanism takes the **Lévy–Khintchine form**

$$\psi(u) = \alpha u + \beta u^2 + \int_{(0,\infty)} (e^{-ur} - 1 + ur) \pi(dr),$$

which decomposes naturally into:

- a linear drift/killing component;
- a quadratic Brownian fluctuation component;
- a jump component determined by a Lévy measure.

One of the main themes of this thesis is that these three components already appear at the discrete Galton–Watson level. In Le Gall’s approximation scheme, the branching events are decomposed into:

- killing without descendants;
- binary branching;
- Poisson-type jump branching.

After suitable rescaling, these discrete mechanisms converge respectively to the three terms in the Lévy–Khintchine branching mechanism.

Special attention is devoted to the **stable branching case**

$$\psi(u) = \kappa u^\alpha, \quad 1 < \alpha \leq 2.$$

When $1 < \alpha < 2$, the limiting process is purely jump-driven and arises from the Lévy jump component. When $\alpha = 2$, the model reduces to the classical Brownian/Feller branching diffusion.

Finally, we consider the extension from pure branching systems to **continuous-state branching processes with immigration (CBIs)**. By adding immigration events to the discrete Galton–Watson model and imposing suitable scaling assumptions on the immigration mechanism, we obtain a limiting immigration drift term of the form $c_t dt$. In this sense, the deterministic term $c_t dt$ appearing in the stable-driven equation can be interpreted as the continuum scaling limit of immigration in the underlying discrete branching system.

Overall, this thesis combines three closely related perspectives:

- the probabilistic scaling limit from Galton–Watson processes to CSBPs;
- the role of stable Lévy fluctuations in the infinite-variance regime;

- the structural interpretation of immigration through continuous-state branching processes with immigration.

The purpose of this work is not to develop a complete theory of stable CBIs, but rather to clarify the relationship between these viewpoints and to provide a preliminary analysis of the corresponding scaling structures, self-similarity properties, and Laplace transform formulations.

2. Continuous-State Branching Processes

2.1. Definition

A continuous-state branching process (CSBP) can be viewed as the continuous-state analogue of a Galton–Watson branching process. This viewpoint is standard in the theory of branching processes: Galton–Watson processes describe discrete populations, while CSBPs describe branching populations whose state space is continuous [1, 2]. Its defining feature is not a specific stochastic differential equation, but rather the **branching property**.

Definition 2.1. A Markov process $X = (X_t)_{t \geq 0}$ taking values in $[0, \infty)$ is called a continuous-state branching process (CSBP) if, for any $x, y \geq 0$ and any $t \geq 0$, one has

$$X_t^{(x+y)} \stackrel{d}{=} X_t^{(x)} + \tilde{X}_t^{(y)},$$

where the two terms on the right-hand side are independent, and $X^{(x)}$ denotes the process starting from initial value x [1].

Remark 2.2. The stochastic differential equation

$$dY_t = 2\sqrt{Y_t} dW_t + \delta dt$$

describes a squared Bessel process, where the term δdt represents a drift depending on the dimension parameter δ . In particular, when $\delta = 0$, the process Y_t corresponds to the 0-dimensional squared Bessel process. This is closely related to the Feller branching diffusion, which is a basic example of a CSBP with quadratic branching mechanism [1].

2.2. Laplace Transform Characterization

The branching property of a CSBP is equivalent to an exponential form of its Laplace transform. More precisely, for any $\lambda \geq 0$, one has

$$\mathbb{E}_{x+y}(e^{-\lambda X_t}) = \mathbb{E}_x(e^{-\lambda X_t}) \mathbb{E}_y(e^{-\lambda X_t}).$$

As a consequence, there exists a function $u_t(\lambda)$ such that

$$\mathbb{E}_x(e^{-\lambda X_t}) = \exp(-x u_t(\lambda)).$$

This exponential-affine form is the standard Laplace transform characterization of CSBPs [1].

Remark 2.3. The multiplicative structure of the Laplace transform is a direct consequence of the branching property. It reflects the fact that the evolution from total initial mass $x + y$ can be decomposed into two independent contributions corresponding to the initial masses x and y .

3. Stable Distributions and Stable Lévy Processes

3.1. Stable Lévy Processes

Definition 3.1. A random variable Z_α is called an α -stable random variable if its distribution belongs to the class of stable laws. In the symmetric case, its characteristic function can be written as

$$\mathbb{E}(e^{i\theta Z_\alpha}) = \exp(-c|\theta|^\alpha), \quad 0 < \alpha \leq 2,$$

for some constant $c > 0$ [1, 3].

Definition 3.2. A Lévy process $Y = (Y_t)_{t \geq 0}$ is called an α -stable Lévy process if

$$Y_1 \stackrel{d}{=} Z_\alpha,$$

where Z_α is an α -stable random variable [1].

By the stationary and independent increments of Lévy processes, we obtain:

Proposition 3.3. Let $Y = (Y_t)_{t \geq 0}$ be an α -stable Lévy process. Then for any $t > 0$,

$$Y_t \stackrel{d}{=} t^{1/\alpha} Y_1,$$

and for any $0 \leq s < t$,

$$Y_t - Y_s \stackrel{d}{=} (t - s)^{1/\alpha} Y_1.$$

3.2. Self-Similarity of Stable Lévy Processes

Definition 3.4. A stochastic process $X = (X_t)_{t \geq 0}$ is said to be self-similar with index $H > 0$ if, for any $c > 0$,

$$(X_{ct})_{t \geq 0} \stackrel{d}{=} (c^H X_t)_{t \geq 0}.$$

Proposition 3.5. Every α -stable Lévy process is self-similar with self-similarity index $1/\alpha$ [1].

Remark 3.6. This implication is one-directional: every α -stable Lévy process is self-similar, but the converse does not hold in general.

3.3. Brownian Motion as the 2-Stable Case

Brownian motion B_t satisfies

$$B_t \stackrel{d}{=} t^{1/2} B_1,$$

hence it is a special case of a 2-stable Lévy process [1].

Remark 3.7. More precisely, Brownian motion is a 2-stable Lévy process. However, a general Gaussian process need not be a Lévy process, since it may fail to have stationary and independent increments.

3.4. Domain of Attraction of Stable Laws

Definition 3.8 (Domain of attraction of stable laws). Let ξ be a random variable. If there exist sequences of constants (a_n) and (b_n) such that

$$\frac{1}{a_n} \left(\sum_{i=1}^n \xi_i - b_n \right) \Rightarrow Z_\alpha,$$

where Z_α is an α -stable random variable ($0 < \alpha \leq 2$), then the distribution of ξ is said to belong to the domain of attraction of an α -stable law.

Theorem 3.9 (Regular variation criterion). Let $1 < \alpha < 2$. If a random variable ξ satisfies the tail condition

$$\mathbb{P}(|\xi| > x) \sim x^{-\alpha} L(x), \quad x \rightarrow \infty,$$

where $L(x)$ is a slowly varying function, i.e.

$$\lim_{x \rightarrow \infty} \frac{L(cx)}{L(x)} = 1, \quad \forall c > 0,$$

then the distribution of ξ belongs to the domain of attraction of an α -stable law [1].

Proposition 3.10 (Equivalent tail characterization). Let $1 < \alpha < 2$. If there exist constants $c_+, c_- \geq 0$ such that

$$\mathbb{P}(\xi > x) \sim c_+ x^{-\alpha} L(x), \quad \mathbb{P}(\xi < -x) \sim c_- x^{-\alpha} L(x),$$

then $\xi \in \mathcal{D}(\alpha\text{-stable})$, and the skewness of the limiting stable law is determined by c_+ and c_- [1].

Remark 3.11. It is important to emphasize that having infinite variance does not imply belonging to the domain of attraction of a stable law.

4. Infinite Divisibility and the Lévy–Khintchine Formula

Here we explain the structural relationship between infinitely divisible distributions, Lévy processes, and continuous-state branching processes (CSBPs).

The key point is that infinitely divisible distributions admit characteristic functions of exponential form, whose exponents necessarily have the Lévy–Khintchine structure. This representation extends naturally to Lévy processes through stationary and independent increments, and later reappears in CSBPs through the branching mechanism governing their Laplace transforms.

Thus, the same Lévy–Khintchine structure underlies:

$$\text{infinitely divisible distributions} \rightarrow \text{Lévy processes} \rightarrow \text{CSBP branching mechanisms}.$$

4.1. Structure of Infinitely Divisible Distributions

Theorem 4.1 (Lévy–Khintchine representation for infinitely divisible distributions [3, Chapter 3]). Let μ be an infinitely divisible probability measure on $\mathbb{R}$, and let

$$\widehat{\mu}(\theta) = \int_{\mathbb{R}} e^{i\theta x} \mu(dx)$$

be its characteristic function.

Then there exist:

- a constant $b \in \mathbb{R}$;
- a constant $\sigma^2 \geq 0$;
- a measure ν on $\mathbb{R} \setminus \{0\}$ satisfying

$$\int_{\mathbb{R}} (1 \wedge x^2) \nu(dx) < \infty,$$

such that

$$\widehat{\mu}(\theta) = \exp(\Psi(\theta)),$$

where the characteristic exponent Ψ is given by

$$\Psi(\theta) = ib\theta - \frac{1}{2}\sigma^2\theta^2 + \int_{\mathbb{R}} (e^{i\theta x} - 1 - i\theta x \mathbf{1}_{\{|x|<1\}}) \nu(dx).$$

The triple (b, σ^2, ν) is uniquely determined by μ .

Conversely, every function of the above form is the characteristic function of an infinitely divisible probability distribution.

Remark 4.2. The three terms in the Lévy–Khintchine representation correspond respectively

to: deterministic drift, Brownian fluctuation and jump component. **Therefore, the structure of an infinitely divisible process is completely determined by these three mechanisms.**

4.2. From Infinitely Divisible Distributions to Lévy Processes

The Lévy–Khintchine representation extends naturally from infinitely divisible distributions to Lévy processes. If $(X_t)_{t \geq 0}$ is a Lévy process, then for every $t \geq 0$,

$$\mathbb{E}[e^{i\theta X_t}] = \exp\{t\Psi(\theta)\},$$

where Ψ is the Lévy–Khintchine exponent associated with the infinitely divisible distribution of X_1 .

Hence the entire law of a Lévy process is completely determined by the Lévy triplet

$$(b, \sigma^2, \nu),$$

corresponding respectively to drift or killing, Brownian fluctuation and jump mechanism.

A closely related structure appears in continuous-state branching processes (CSBPs). Instead of characteristic functions, CSBPs are characterized through Laplace transforms. More precisely, if $(Y_t)_{t \geq 0}$ is a CSBP, then

$$\mathbb{E}_x[e^{-\lambda Y_t}] = \exp\{-xu_t(\lambda)\},$$

where $u_t(\lambda)$ satisfies

$$u_t(\lambda) + \int_0^t \psi(u_s(\lambda)) ds = \lambda.$$

The branching mechanism ψ necessarily takes the Lévy–Khintchine form

$$\psi(u) = \alpha u + \beta u^2 + \int_{(0, \infty)} (e^{-ur} - 1 + ur) \pi(dr),$$

where:

$$\alpha \geq 0, \quad \beta \geq 0,$$

and π is a measure satisfying

$$\int_{(0, \infty)} (r \wedge r^2) \pi(dr) < \infty.$$

Therefore, just as **Lévy processes are determined by their Lévy–Khintchine exponents**, CS-

BPs are completely determined by their branching mechanisms. In particular, the branching structure of a CSBP is governed by the same three components:

drift/killing, Brownian branching, jump branching.

5. Feller's Scaling Limit Theorem and the Role of the Central Limit Theorem

5.1. Heuristic Statement of Feller's Theorem

Let $(Z_n)_{n \geq 0}$ be a critical Galton–Watson process satisfying

$$\mathbb{E}(Z_1) = 1, \quad \text{Var}(Z_1) = \sigma^2 < \infty.$$

Consider the rescaled process

$$X_t^{(n)} := \frac{1}{n} Z_{\lfloor nt \rfloor}.$$

Under suitable assumptions on the initial condition, for example

$$\frac{Z_0}{n} \rightarrow x \geq 0,$$

the rescaled processes converge in distribution to a limiting process X :

$$X^{(n)} \Rightarrow X.$$

The limiting process is the Feller branching diffusion, which satisfies the stochastic differential equation

$$dX_t = \sigma \sqrt{X_t} dW_t, \quad X_0 = x$$

[1, 2].

Remark 5.1. The Feller branching diffusion

$$dX_t = \sigma \sqrt{X_t} dW_t, \quad X_0 = x \geq 0,$$

is one of the most basic examples of a continuous-state branching process (CSBP) [1].

Moreover, it is closely related to the 0-dimensional squared Bessel process. Indeed, if we define

$$R_t := \frac{4}{\sigma^2} X_t,$$

then, by Itô's formula,

$$dR_t = 2\sqrt{R_t} dW_t.$$

This is the stochastic differential equation of a BESQ^0 process [1].

Proposition 5.2. After a linear change of scale, the Feller branching diffusion is equivalent to a 0-dimensional squared Bessel process [1].

5.2. When the Central Limit Theorem Fails: The Infinite Variance Case

In the finite variance setting, Feller's scaling limit theorem is closely related to the classical Central Limit Theorem. The Brownian noise in the limiting diffusion arises from Gaussian fluctuations of the offspring distribution under suitable scaling [3].

We now consider the case where the offspring distribution has infinite variance and belongs to the domain of attraction of a stable law. In this heavy-tailed regime, the classical Central Limit Theorem is replaced by a stable limit theorem, and the Gaussian fluctuation is expected to be replaced by an α -stable fluctuation [3].

Theorem 5.3 (Classical stable limit theorem). Let $(\xi_i)_{i \geq 1}$ be a sequence of i.i.d. random variables. Suppose that the distribution of ξ_i belongs to the domain of attraction of an α -stable law. Then there exists a normalizing sequence (a_n) , regularly varying with index $1/\alpha$, such that

$$a_n \sim n^{1/\alpha} L(n),$$

where L is slowly varying, and

$$\frac{1}{a_n} \sum_{i=1}^n \xi_i \Rightarrow Z_\alpha,$$

where Z_α is an α -stable random variable [3].

This theorem shows that, in the infinite variance setting, Gaussian fluctuations are replaced by α -stable fluctuations. Consequently, the natural normalization is no longer $\sqrt{n}$, but rather $n^{1/\alpha}$, possibly with a slowly varying correction [3].

Proposition 5.4 (Heuristic stable scaling limit for branching processes). Let $(Z_n)_{n \geq 0}$ be a critical Galton–Watson process satisfying

$$\mathbb{E}(Z_1) = 1, \quad \text{Var}(Z_1) = \infty.$$

Assume further that the offspring distribution belongs to the domain of attraction of an α -stable law, where $1 < \alpha \leq 2$.

Suppose that there exists a normalizing sequence (a_n) with

$$a_n \sim n^{1/\alpha} L(n),$$

such that the rescaled process

$$X_t^{(n)} := \frac{1}{a_n} Z_{\lfloor nt \rfloor}$$

converges in distribution to a limiting process Y_t .

Then, at a heuristic level, the limiting process is expected to be driven by an α -stable Lévy process. In particular, it is natural to consider a stable-driven stochastic differential equation of the form

$$dY_t = a Y_t^{\frac{1}{1+\beta}} d\beta_t,$$

where

- β_t is an α -stable Lévy process with $\alpha = 1 + \beta$;
- the constant $a > 0$ depends on the normalization and on the scale of the offspring distribution.

This candidate limiting process is expected to satisfy a self-similar scaling relation. More precisely, its natural self-similarity index is

$$H = \frac{1}{\alpha - 1} = \frac{1}{\beta},$$

so that, formally,

$$(Y_{ct})_{t \geq 0} \stackrel{d}{=} (c^H Y_t)_{t \geq 0}.$$

Remark 5.5. The stable-driven model above should be understood as a heuristic candidate for the heavy-tailed scaling limit. In general, such a model does not automatically satisfy the branching property.

In this thesis, rather than characterizing all conditions under which the process becomes a CSBP, we restrict our attention to the class of processes that do satisfy the branching property. That is, we work under the assumption that the limiting process is a continuous-state branching process, and focus on the corresponding structure under this modeling choice.

6. Limit of Galton–Watson processes to CSBPs (with Immigration)

6.1. Preliminaries: Convergence from Galton–Watson Processes

Theorem 6.1 (Scaling limit of Galton–Watson processes to CSBPs). Let $(Z_t^\varepsilon)_{t \geq 0}$ be a continuous-time Galton–Watson process with branching mechanism specified by parameters $(\alpha_\varepsilon, \beta_\varepsilon, \gamma_\varepsilon, \pi_\varepsilon)$ as in Le Gall’s construction [1]. Let $m_\varepsilon \rightarrow \infty$ and define the rescaled process

$$X_t^\varepsilon = \frac{1}{m_\varepsilon} Z_t^\varepsilon.$$

Assume that for every $u \geq 0$,

$$m_\varepsilon \rho_\varepsilon \left(\varphi_\varepsilon(e^{-u/m_\varepsilon}) - e^{-u/m_\varepsilon} \right) \longrightarrow \psi(u),$$

where $\rho_\varepsilon = \alpha_\varepsilon + \beta_\varepsilon + \gamma_\varepsilon$ and

$$\psi(u) = pu + qu^2 + \int_{(0,\infty)} (e^{-ur} - 1 + ur) \pi(dr)$$

is a branching mechanism of Lévy–Khintchine form [1].

Then X^ε converges in distribution to a continuous-state branching process (CSBP) with branching mechanism ψ [1].

Remark 6.2 (Stable branching case). If the offspring distributions belong to the domain of attraction of a stable law with index $\alpha \in (1, 2)$, then the limiting branching mechanism takes the form

$$\psi(u) = \kappa u^\alpha,$$

which corresponds to a purely jump-type CSBP with Lévy measure

$$\pi(dr) = Cr^{-1-\alpha}dr,$$

and no linear or quadratic component (i.e., the drift and diffusion coefficients vanish) [1].

6.2. Extension to Immigration

Proposition 6.3 (Extension to Galton–Watson processes with immigration). Let $(Z_t^\varepsilon)_{t \geq 0}$ be a continuous-time Galton–Watson process with immigration. In addition to the branching mechanism described above, suppose that immigration occurs according to a (possibly time-inhomogeneous) Poisson process with rate $\lambda_\varepsilon(t)$, and that each immigration event introduces a random number of individuals with generating function $h_{\varepsilon,t}$.

Assume that, for every $u \geq 0$,

$$\lambda_\varepsilon(t) \left(1 - h_{\varepsilon,t}(e^{-u/m_\varepsilon}) \right) \longrightarrow \Phi_t(u),$$

where Φ_t is a continuous immigration mechanism.

Then, under standard additional assumptions, the rescaled process

$$X_t^\varepsilon = \frac{1}{m_\varepsilon} Z_t^\varepsilon$$

converges to a continuous-state branching process with immigration (CBI) characterized by branching mechanism ψ and immigration mechanism Φ_t [4, 5].

In particular, if

$$\Phi_t(u) = c_t u,$$

the limiting process admits a representation involving an immigration drift term $c_t dt$ [4].

7. The Current Research Problem

Based on the scaling limit theorem and its extension with immigration discussed above, we consider the following candidate continuous model driven by stable Lévy noise:

$$dY_t = aY_{t-}^{\frac{1}{1+\beta}} d\beta_t + c_t dt,$$

where β_t is a $(1 + \beta)$ -stable spectrally positive Lévy process.

When $c_t \equiv 0$, this model corresponds to a stable continuous-state branching process with branching mechanism

$$\psi(u) = \kappa u^{1+\beta},$$

and can be obtained as the scaling limit of suitably rescaled Galton–Watson processes.

When $c_t \neq 0$, we work under the corresponding CBI framework, where the additional term $c_t dt$ is interpreted as an immigration drift. In this setting, the process is expected to arise as the scaling limit of Galton–Watson processes with immigration.

7.1. Calculation

In this subsection, we answer two preliminary questions concerning the candidate stable-driven model

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t + c_t dt, \quad \alpha = 1 + \beta \in (1, 2],$$

where β_t is a spectrally positive α -stable Lévy process. The constant $a > 0$ depends on the normalization of the stable noise. Throughout this subsection, we assume that $c_t \geq 0$ is deterministic and locally integrable, so that the term $c_t dt$ can be interpreted as a deterministic immigration mechanism.

7.1.1. Self-similarity condition on c_t

We first determine when the process can preserve a self-similar scaling property. Recall that an α -stable Lévy process satisfies

$$(\beta_{rt})_{t \geq 0} \stackrel{d}{=} (r^{1/\alpha} \beta_t)_{t \geq 0}, \quad r > 0.$$

Suppose that Y_t is self-similar with index $H > 0$, in the sense that

$$\tilde{Y}_t := r^{-H} Y_{rt}$$

has the same law as Y_t after the corresponding rescaling of the initial condition.

Using the SDE for Y_{rt} , we compute formally

$$dY_{rt} = aY_{rt-}^{1/\alpha} d\beta_{rt} + c_{rt} r dt.$$

Therefore

$$d\tilde{Y}_t = r^{-H} dY_{rt} = ar^{-H} Y_{rt-}^{1/\alpha} d\beta_{rt} + r^{1-H} c_{rt} dt.$$

Since

$$Y_{rt} = r^H \tilde{Y}_t,$$

and using the scaling property of β_t , we obtain heuristically

$$d\tilde{Y}_t \stackrel{d}{=} ar^{-H+H/\alpha+1/\alpha} \tilde{Y}_{t-}^{1/\alpha} d\beta_t + r^{1-H} c_{rt} dt.$$

For the noise coefficient to remain unchanged, we require

$$-H + \frac{H}{\alpha} + \frac{1}{\alpha} = 0.$$

Equivalently,

$$H(\alpha - 1) = 1,$$

and hence

$$H = \frac{1}{\alpha - 1} = \frac{1}{\beta}.$$

Thus the stable branching part has the natural self-similarity index

$$H = \frac{1}{\alpha - 1}.$$

It remains to require that the deterministic immigration term also preserves the same scaling.

We need

$$r^{1-H} c_{rt} = c_t, \quad \forall r > 0, t > 0.$$

Equivalently,

$$c_{rt} = r^{H-1} c_t.$$

This means that c_t must be a homogeneous function of degree $H - 1$. Therefore there exists a

constant $K \geq 0$ such that

$$c_t = Kt^{H-1}.$$

Since $H = 1/(\alpha - 1)$, this becomes

$$c_t = Kt^{\frac{1}{\alpha-1}-1} = Kt^{\frac{2-\alpha}{\alpha-1}}.$$

Therefore, the model

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t + c_t dt$$

preserves the same self-similar scaling as the stable branching part precisely when the immigration coefficient has the homogeneous form

$$c_t = Kt^{\frac{2-\alpha}{\alpha-1}}$$

for some constant $K \geq 0$.

7.1.2. Laplace transform of Y_t

We next compute the Laplace transform of Y_t for a general deterministic immigration coefficient c_t by its generator.

Let

$$f_\lambda(x) = e^{-\lambda x}, \quad \lambda \geq 0.$$

Assuming that the process admits the CSBP structure with branching mechanism ψ , the generator of the pure branching part satisfies

$$\mathcal{L}_0 e^{-\lambda x} = x\psi(\lambda)e^{-\lambda x}.$$

After adding the deterministic immigration drift $c_t dt$, the time-dependent generator becomes

$$\mathcal{L}_t f(x) = \mathcal{L}_0 f(x) + c_t f'(x).$$

Applying this to $f_\lambda(x) = e^{-\lambda x}$ gives

$$\mathcal{L}_t e^{-\lambda x} = x\psi(\lambda)e^{-\lambda x} - c_t \lambda e^{-\lambda x}.$$

Hence

$$\mathcal{L}_t e^{-\lambda x} = (x\psi(\lambda) - c_t \lambda) e^{-\lambda x}.$$

The affine form of the generator suggests that the Laplace transform should also have an exponential-

affine form. More precisely, for $Y_0 = x$,

$$\mathbb{E}_x [e^{-\lambda Y_t}] = \exp \left\{ -xv_t(\lambda) - \int_0^t c_s v_{t-s}(\lambda) ds \right\},$$

where $v_t(\lambda)$ solves

$$\frac{d}{dt}v_t(\lambda) = -\kappa v_t(\lambda)^\alpha, \quad v_0(\lambda) = \lambda.$$

Solving this ODE gives

$$v_t(\lambda) = \left(\lambda^{1-\alpha} + \kappa(\alpha-1)t \right)^{-\frac{1}{\alpha-1}}.$$

7.2. The Limiting Problem

We now formulate the limiting problem following Le Gall's construction of continuous-time Galton–Watson approximations. The main point of this subsection is first to recall the case without immigration, and then to explain how the same limiting structure can be extended when an immigration term is added.

7.2.1. The case without immigration

Recall that a continuous-state branching process is characterized by a branching mechanism of Lévy–Khintchine type

$$\psi(u) = \alpha u + \beta u^2 + \int_{(0,\infty)} (e^{-ur} - 1 + ur) \pi(dr), \quad u \geq 0,$$

where $\alpha \geq 0$, $\beta \geq 0$, and π is a σ -finite measure on $(0, \infty)$ satisfying

$$\int_{(0,\infty)} (r \wedge r^2) \pi(dr) < \infty.$$

This formula shows that, in the critical or subcritical CSBP setting, there are **only three possible branching mechanisms**:

- (1) the linear term αu , corresponding to killing or deterministic linear drift;
- (2) the quadratic term βu^2 , corresponding to Brownian fluctuation or Feller diffusion;
- (3) the jump term described by the Lévy measure π , corresponding to sudden positive jumps.

Hence Le Gall's approximation scheme is representative: it constructs a continuous-time Galton–Watson process whose individuals reproduce exactly through these three types of elementary events.

Let $X^\varepsilon = (X_t^\varepsilon, t \geq 0)$ be a continuous-time Galton–Watson process. Then we can assume

individuals die at rate

$$\rho_\varepsilon = \alpha_\varepsilon + \beta_\varepsilon + \gamma_\varepsilon.$$

When an individual dies, one of the following three events occurs:

- with probability $\alpha_\varepsilon/\rho_\varepsilon$, the individual dies without descendants;
- with probability $\beta_\varepsilon/\rho_\varepsilon$, the individual gives rise to 0 or 2 children with probability 1/2 each;
- with probability $\gamma_\varepsilon/\rho_\varepsilon$, the individual gives rise to a random number of offspring. More precisely, let V have distribution

$$\pi_\varepsilon(dv) = \pi((\varepsilon, \infty))^{-1} \mathbf{1}_{\{v > \varepsilon\}} \pi(dv),$$

and, conditionally on V , let the number of offspring be Poisson with parameter $m_\varepsilon V$.

The offspring generating function is therefore:

$$\varphi_\varepsilon(r) = \frac{\alpha_\varepsilon}{\rho_\varepsilon} + \frac{\beta_\varepsilon}{\rho_\varepsilon} \frac{1+r^2}{2} + \frac{\gamma_\varepsilon}{\rho_\varepsilon} \int_{(0, \infty)} e^{-m_\varepsilon v(1-r)} \pi_\varepsilon(dv).$$

Define:

$$g_\varepsilon(r) = \rho_\varepsilon(\varphi_\varepsilon(r) - r).$$

For $r \in [0, 1]$, we let

$$v_t^\varepsilon(r) = \mathbf{E}_1[r^{X_t^\varepsilon}],$$

then the standard generating function equation for continuous-time Galton–Watson processes gives

$$\boxed{v_t^\varepsilon(r) = r + \int_0^t g_\varepsilon(v_s^\varepsilon(r)) ds.} \tag{1}$$

Equation (1) is the fundamental generating-function equation for the continuous-time branching process.

We are interested in the scaling limit of

$$\frac{1}{m_\varepsilon} X_t^\varepsilon, \quad m_\varepsilon \rightarrow \infty,$$

when the initial condition is

$$X_0^\varepsilon = [m_\varepsilon x].$$

For $\lambda \geq 0$,

$$\mathbf{E}_{[m_\varepsilon x]} [\exp(-\lambda m_\varepsilon^{-1} X_t^\varepsilon)] = (v_t^\varepsilon(e^{-\lambda/m_\varepsilon}))^{[m_\varepsilon x]}.$$

This motivates the definition

$$\boxed{u_t^\varepsilon(\lambda) = m_\varepsilon \left(1 - v_t^\varepsilon(e^{-\lambda/m_\varepsilon})\right)}. \quad (2)$$

By the definition of u_t^ε , we have

$$v_t^\varepsilon(e^{-\lambda/m_\varepsilon}) = 1 - \frac{u_t^\varepsilon(\lambda)}{m_\varepsilon}.$$

Taking

$$r = e^{-\lambda/m_\varepsilon}$$

in (1), we obtain

$$1 - \frac{u_t^\varepsilon(\lambda)}{m_\varepsilon} = e^{-\lambda/m_\varepsilon} + \int_0^t g_\varepsilon \left(1 - \frac{u_s^\varepsilon(\lambda)}{m_\varepsilon}\right) ds.$$

Rearranging this identity and multiplying by m_ε gives

$$u_t^\varepsilon(\lambda) + \int_0^t m_\varepsilon g_\varepsilon \left(1 - \frac{u_s^\varepsilon(\lambda)}{m_\varepsilon}\right) ds = m_\varepsilon(1 - e^{-\lambda/m_\varepsilon}).$$

Therefore, defining

$$\psi_\varepsilon(u) = m_\varepsilon g_\varepsilon(1 - m_\varepsilon^{-1}u),$$

we conclude that u_t^ε satisfies

$$\boxed{u_t^\varepsilon(\lambda) + \int_0^t \psi_\varepsilon(u_s^\varepsilon(\lambda)) ds = m_\varepsilon(1 - e^{-\lambda/m_\varepsilon}).} \quad (3)$$

Equation (3) is the rescaled Laplace equation corresponding to the branching process. In the scaling limit, it converges to the defining integral equation of a CSBP.

A direct computation gives

$$\boxed{\begin{aligned} \psi_\varepsilon(u) = & \left(\alpha_\varepsilon - m_\varepsilon \gamma_\varepsilon \int_{(0,\infty)} r \pi_\varepsilon(dr) + \gamma_\varepsilon \right) u \\ & + \frac{m_\varepsilon^{-1} \beta_\varepsilon}{2} u^2 \\ & + m_\varepsilon \gamma_\varepsilon \int_{(0,\infty)} (e^{-ur} - 1 + ur) \pi_\varepsilon(dr). \end{aligned}} \quad (4)$$

Equation (4) already exhibits the three components of the Lévy–Khintchine branching mechanism:

- the linear drift/killing term;
- the quadratic Brownian fluctuation term;
- the jump term determined by the Lévy measure.

Hence ψ_ε can be viewed as the **discrete approximation of limiting branching mechanism** ψ .

Following Le Gall, we choose the parameters so that

$$m_\varepsilon \rightarrow \infty, \tag{i}$$

$$m_\varepsilon \gamma_\varepsilon \pi((\varepsilon, \infty))^{-1} \rightarrow 1 \quad \text{if } \pi \neq 0, \quad \gamma_\varepsilon = 0 \quad \text{if } \pi = 0, \tag{ii}$$

$$\frac{1}{2} m_\varepsilon^{-1} \beta_\varepsilon \rightarrow \beta, \tag{iii}$$

and

$$\alpha_\varepsilon - m_\varepsilon \gamma_\varepsilon \int_{(0, \infty)} r \pi_\varepsilon(dr) + \gamma_\varepsilon \rightarrow \alpha, \tag{iv}$$

with

$$\alpha_\varepsilon - m_\varepsilon \gamma_\varepsilon \int_{(0, \infty)} r \pi_\varepsilon(dr) + \gamma_\varepsilon \geq 0$$

for every $\varepsilon > 0$. Under these assumptions,

$$\psi_\varepsilon(u) \rightarrow \psi(u)$$

uniformly on compact subsets of $\mathbb{R}_+$.

We now state the resulting approximation proposition.

Proposition 7.1. Suppose that conditions (i)–(iv) hold. Then, for every $t > 0$ and $x \geq 0$, the law of $m_\varepsilon^{-1} X_t^\varepsilon$ under $\mathbf{P}_{[m_\varepsilon x]}$ converges, as $\varepsilon \downarrow 0$, to a probability measure $P_t(x, dy)$. Moreover, the kernels $P_t(x, dy)$ define the CSBP associated with the branching mechanism ψ .

Proof. Let $u_t(\lambda)$ be the unique nonnegative solution of

$$u_t(\lambda) + \int_0^t \psi(u_s(\lambda)) ds = \lambda.$$

Since $\psi_\varepsilon \rightarrow \psi$ uniformly on compact subsets of $\mathbb{R}_+$ and ψ is locally Lipschitz on bounded intervals away from possible irrelevant boundary singularities, comparing the integral equations for u_t^ε and u_t gives

$$|u_t^\varepsilon(\lambda) - u_t(\lambda)| \leq C_\lambda \int_0^t |u_s^\varepsilon(\lambda) - u_s(\lambda)| ds + a(\varepsilon, \lambda),$$

where $a(\varepsilon, \lambda) \rightarrow 0$ as $\varepsilon \downarrow 0$. By Gronwall's lemma,

$$u_t^\varepsilon(\lambda) \rightarrow u_t(\lambda)$$

uniformly for t in compact intervals.

Therefore,

$$\begin{aligned} \mathbf{E}_{[m_\varepsilon x]} \left[e^{-\lambda m_\varepsilon^{-1} X_t^\varepsilon} \right] &= \left(1 - \frac{u_t^\varepsilon(\lambda)}{m_\varepsilon} \right)^{[m_\varepsilon x]} \\ &\longrightarrow \exp\{-x u_t(\lambda)\}. \end{aligned}$$

By the continuity theorem for Laplace transforms, this determines a probability measure $P_t(x, dy)$ satisfying

$$\int_{\mathbb{R}_+} e^{-\lambda y} P_t(x, dy) = e^{-x u_t(\lambda)}.$$

The Chapman–Kolmogorov relation follows from the semigroup identity

$$u_{t+s} = u_t \circ u_s,$$

which is obtained from the integral equation. Finally, the branching property follows from

$$e^{-(x+x')u_t(\lambda)} = e^{-xu_t(\lambda)} e^{-x'u_t(\lambda)}.$$

Hence

$$P_t(x + x', \cdot) = P_t(x, \cdot) * P_t(x', \cdot),$$

so the limiting process is the CSBP with branching mechanism ψ . □

Remark 7.2. Conditions (i)–(iv) are sufficient conditions ensuring that

$$\psi_\varepsilon(u) \rightarrow \psi(u),$$

hence the rescaled Galton–Watson processes converge to the CSBP with the branching mechanism ψ . They are not intended to be necessary conditions, since different choices of truncation, normalization, or compensation may lead to the same limiting branching mechanism.

Remark 7.3. Nevertheless, these assumptions are canonical up to normalization and time scaling. They provide a **discrete realization of the Lévy–Khintchine decomposition of the limiting branching mechanism**. More precisely:

- condition (i) corresponds to the continuum mass scaling;
- condition (ii) normalizes the jump intensity associated with the Lévy measure;

- condition (iii) produces the quadratic Brownian fluctuation term;
- condition (iv) provides the drift compensation required by the truncation of the jump part.

7.2.2. The stable branching case

We now apply the preceding construction to the stable mechanism

$$\psi(u) = \kappa u^\alpha, \quad 1 < \alpha \leq 2.$$

First suppose that $1 < \alpha < 2$. In this case the target mechanism is purely of jump type. We take

$$\pi(dv) = C_\alpha v^{-1-\alpha} dv, \quad v > 0,$$

where $C_\alpha > 0$ is chosen so that

$$\kappa u^\alpha = \int_0^\infty (e^{-uv} - 1 + uv) \pi(dv).$$

There is no Brownian part, so we choose

$$\beta_\varepsilon = 0.$$

Following Le Gall's truncation of the jump measure, set

$$\pi_\varepsilon(dv) = \pi((\varepsilon, \infty))^{-1} \mathbf{1}_{\{v > \varepsilon\}} \pi(dv),$$

and choose

$$\gamma_\varepsilon = \frac{1}{m_\varepsilon} \pi((\varepsilon, \infty)).$$

Then

$$m_\varepsilon \gamma_\varepsilon \pi((\varepsilon, \infty))^{-1} = 1.$$

The linear compensation is chosen by setting

$$\alpha_\varepsilon = m_\varepsilon \gamma_\varepsilon \int_{(0, \infty)} v \pi_\varepsilon(dv) = \int_\varepsilon^\infty v \pi(dv).$$

With this choice, the linear term produced by the truncated jump part is exactly compensated, and the limit is

$$\psi_\varepsilon(u) \rightarrow \int_0^\infty (e^{-uv} - 1 + uv) \pi(dv) = \kappa u^\alpha.$$

Thus, for $1 < \alpha < 2$, the stable CSBP is obtained from the jump component in Le Gall's construction.

When $\alpha = 2$, the target mechanism is quadratic:

$$\psi(u) = \kappa u^2.$$

This is the Brownian or Feller diffusion case. We therefore suppress the killing and jump components by taking

$$\alpha_\varepsilon = 0, \quad \gamma_\varepsilon = 0, \quad \pi_\varepsilon = 0,$$

and choose

$$\beta_\varepsilon = 2\kappa m_\varepsilon.$$

Indeed, since

$$1 - e^{-u/m_\varepsilon} \sim \frac{u}{m_\varepsilon},$$

we have

$$m_\varepsilon \beta_\varepsilon \left(\frac{1 + e^{-2u/m_\varepsilon}}{2} - e^{-u/m_\varepsilon} \right) = m_\varepsilon \beta_\varepsilon \frac{(1 - e^{-u/m_\varepsilon})^2}{2} \rightarrow \kappa u^2.$$

Hence the Brownian case $\alpha = 2$ is recovered from the quadratic component in Le Gall's approximation.

7.2.3. Extension to immigration

The preceding construction treats the pure branching case. To incorporate immigration, we add external arrivals to the Galton–Watson population. Let immigration events occur at rate $\lambda_\varepsilon(t)$, and suppose that each immigration event introduces a random number of immigrants with generating function $h_{\varepsilon,t}$.

Under the same mass scaling m_ε^{-1} , the immigration contribution has a nontrivial limit if

$$\lambda_\varepsilon(t) (1 - h_{\varepsilon,t}(e^{-u/m_\varepsilon})) \rightarrow \Phi_t(u), \quad u \geq 0.$$

For the deterministic immigration drift considered in this thesis, the target immigration mechanism is

$$\Phi_t(u) = c_t u.$$

Thus it is enough to impose

$$\lambda_\varepsilon(t) (1 - h_{\varepsilon,t}(e^{-u/m_\varepsilon})) \rightarrow c_t u.$$

For example, if the immigration size $\eta_{\varepsilon,t}$ has uniformly bounded finite mean, then

$$1 - h_{\varepsilon,t}(e^{-u/m_\varepsilon}) \sim \frac{u}{m_\varepsilon} \mathbf{E}[\eta_{\varepsilon,t}],$$

and the required condition becomes

$$\lambda_\varepsilon(t) \frac{\mathbf{E}[\eta_{\varepsilon,t}]}{m_\varepsilon} \rightarrow c_t.$$

A particularly simple realization is the deterministic one-particle immigration case:

$$\eta_{\varepsilon,t} \equiv 1, \quad h_{\varepsilon,t}(r) = r, \quad \lambda_\varepsilon(t) = c_t m_\varepsilon.$$

Then

$$\lambda_\varepsilon(t) (1 - h_{\varepsilon,t}(e^{-u/m_\varepsilon})) = c_t m_\varepsilon (1 - e^{-u/m_\varepsilon}) \rightarrow c_t u.$$

Consequently, the limiting process is expected to be a continuous-state branching process with immigration, whose Laplace transform has the affine form

$$\mathbf{E}_x[e^{-\lambda Y_t}] = \exp \left\{ -x u_t(\lambda) - \int_0^t \Phi_s(u_{t-s}(\lambda)) ds \right\}.$$

In the present case $\Phi_s(u) = c_s u$, so

$$\mathbf{E}_x[e^{-\lambda Y_t}] = \exp \left\{ -x u_t(\lambda) - \int_0^t c_s u_{t-s}(\lambda) ds \right\},$$

where

$$u_t(\lambda) + \int_0^t \psi(u_s(\lambda)) ds = \lambda.$$

For the stable mechanism $\psi(u) = \kappa u^\alpha$, this becomes

$$\frac{d}{dt} u_t(\lambda) = -\kappa u_t(\lambda)^\alpha, \quad u_0(\lambda) = \lambda.$$

Hence

$$u_t(\lambda) = (\lambda^{1-\alpha} + \kappa(\alpha-1)t)^{-\frac{1}{\alpha-1}}.$$

Therefore, the pure Le Gall approximation gives the CSBP part, while the additional convergence condition on the immigration generating functions produces the immigration drift $c_t dt$. In this sense, the stable CBI model

$$dY_t = aY_{t-}^{1/\alpha} d\beta_t + c_t dt, \quad 1 < \alpha \leq 2,$$

can be regarded as the immigration extension of the stable CSBP scaling limit.

7.2.4. Summary

In this subsection, we revisited Le Gall's discrete approximation scheme for continuous-state branching processes and explained how the limiting branching mechanism arises from suitably rescaled continuous-time Galton–Watson processes.

The key point is that the discrete branching events already encode the three components appearing in the Lévy–Khintchine representation of a CSBP branching mechanism:

$$\psi(u) = \alpha u + \beta u^2 + \int_{(0,\infty)} (e^{-ur} - 1 + ur) \pi(dr).$$

More precisely:

- the killing mechanism produces the linear drift term;
- binary branching produces the quadratic Brownian fluctuation term;
- the Poisson offspring mechanism produces the Lévy jump term.

After introducing the rescaled Laplace transform

$$u_t^\varepsilon(\lambda) = m_\varepsilon (1 - v_t^\varepsilon(e^{-\lambda/m_\varepsilon})),$$

the generating-function equation of the Galton–Watson process converges to the integral equation characterizing a CSBP. Under conditions (i)–(iv), the discrete branching mechanisms ψ_ε converge to a limiting branching mechanism ψ of Lévy–Khintchine type.

Applying this construction to the stable case:

$$\psi(u) = \kappa u^\alpha, \quad 1 < \alpha \leq 2,$$

we obtain two limiting regimes:

- for $1 < \alpha < 2$, the limiting CSBP is purely jump-driven and arises from the Lévy jump component;
- for $\alpha = 2$, the limiting process reduces to the Brownian/Feller branching diffusion.

Finally, by adding immigration events to the discrete Galton–Watson model and imposing a compatible scaling condition on the immigration generating functions, the limiting structure extends naturally from CSBPs to continuous-state branching processes with immigration (CBIs). In this framework, the deterministic drift term $c_t dt$ appears as the scaling limit of the immigration mechanism.

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